

Python: exercises on iteration

This notebook was generated in **pseudocode** mode.

Exercise 1: Print Numbers from 1 to N

Exercise Description

Write a function that prints all numbers from 1 to N using a for loop and range.

Your solution

```
def print_numbers(n):  
    # Step 1: Set up a for loop to iterate from 1 to n (inclusive)  
    # Step 2: For each number in the range, print it
```

Test your solution

```
# Test case 1: Test with small number  
print_numbers(5)
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 2: Sum of First N Natural Numbers

Exercise Description

Write a function to calculate the sum of the first N natural numbers ($1 + 2 + 3 + \dots + N$) using a while loop.

Your solution

```
def sum_natural_numbers(n):  
    # Step 1: Initialize sum variable to 0 and counter to 1  
    # Step 2: Set up while loop that continues while counter <= n  
    # Step 3: Add current counter value to sum  
    # Step 4: Increment counter by 1  
    # Step 5: Return the final sum
```

Test your solution

```
# Test case 2: Test with another number
result = sum_natural_numbers(3)
expected = 1 + 2 + 3 # 6
assert result == expected, f"Expected {expected}, got {result}"
print(f"✓ sum_natural_numbers(3) = {result}")
```

Test your solution

```
# Test case: Example from solution
result = sum_natural_numbers(5)
print(f"Sum of first 5 natural numbers: {result}") # 15
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 3: Print Even Numbers from 2 to 20

Exercise Description

Write a function that prints all even numbers between 2 and 20 using a for loop and range.

Your solution

```
def print_even_numbers():  
    # Step 1: Set up a for loop with start=2, stop=21, step=2  
    # Step 2: For each even number in the range, print it
```

Test your solution

```
# Test case 2: Verify range and step  
# Expected: prints 2, 4, 6, 8, 10, 12, 14, 16, 18, 20 (each on new line)  
print_even_numbers()
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 4: Count Down from N to 1 (If Even)

Exercise Description

Write a function to print numbers from N to 1 using a while loop, but only if N is even.

Your solution

```
def countdown_if_even(n):  
    # Step 1: Check if n is even using modulo operator  
    # Step 2: If n is even, set up while loop that continues while n > 0  
    # Step 3: Print current n value  
    # Step 4: Decrement n by 1  
    # Step 5: If n is odd, do nothing
```

Test your solution

```
# Test case 2: Test with odd number (should print nothing)  
countdown_if_even(3)
```

Test your solution

```
# Test case 2 (assert)  
countdown_if_even(6) # Prints 6, 5, 4, 3, 2, 1
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 5: Multiplication Table of 3

Exercise Description

Write a function that prints the multiplication table of 3 (from 3x1 to 3x10) using a for loop.

Your solution

```
def multiplication_table_3():  
    # Step 1: Set up a for loop to iterate from 1 to 10  
    # Step 2: For each multiplier, calculate 3 * multiplier  
    # Step 3: Print the multiplication equation and result
```

Test your solution

```
# Test case 2: Verify format  
# Expected: prints 3 x 1 = 3, 3 x 2 = 6, ..., 3 x 10 = 30  
multiplication_table_3()
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 6: Sum of First N Even Numbers

Exercise Description

Write a function to calculate the sum of the first N even numbers (2 + 4 + 6 + ...) using a while loop.

Your solution

```
def sum_first_n_even(n):  
    # Step 1: Initialize sum to 0, current even number to 2, and count to 0  
    # Step 2: Set up while loop that continues while count < n  
    # Step 3: Add current even number to sum  
    # Step 4: Move to next even number (add 2)  
    # Step 5: Increment count  
    # Step 6: Return the final sum
```

Test your solution

```
# Test case 2: Test with another number  
result = sum_first_n_even(5)  
expected = 2 + 4 + 6 + 8 + 10 # 30  
assert result == expected, f"Expected {expected}, got {result}"  
print(f"✓ sum_first_n_even(5) = {result}")
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 7: Numbers Divisible by 5 (1 to 50)

Exercise Description

Write a function that prints all numbers divisible by 5 between 1 and 50 using a for loop and range.

Your solution

```
def print_divisible_by_5():  
    # Step 1: Set up a for loop with start=5, stop=51, step=5  
    # Step 2: For each number divisible by 5, print it
```

Test your solution

```
# Test case 2: Verify range  
# Expected: prints 5, 10, 15, 20, 25, 30, 35, 40, 45, 50  
print_divisible_by_5()
```


Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 8: Factorial of 5

Exercise Description

Write a function to find the factorial of 5 using a while loop.

Your solution

```
def factorial_5():  
    # Step 1: Initialize factorial to 1 and counter to 5  
    # Step 2: Set up while loop that continues while counter > 0  
    # Step 3: Multiply factorial by current counter value  
    # Step 4: Decrement counter by 1  
    # Step 5: Return the final factorial
```

Test your solution

```
# Test case 2: Verify result
result = factorial_5()
assert result == 120, f"Expected 120, got {result}"
print(f"✓ Factorial of 5 is {result}")
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 9: Sum of Odd Numbers (1 to 20)

Exercise Description

Write a function to calculate the sum of all odd numbers between 1 and 20 using a for loop.

Your solution

```
def sum_odd_numbers():
    # Step 1: Initialize sum to 0
    # Step 2: Set up for loop with start=1, stop=21, step=2 (odd numbers)
```

```
# Step 3: Add each odd number to sum  
# Step 4: Return the final sum
```

Test your solution

```
# Test case 2: Verify result  
result = sum_odd_numbers()  
assert result == 100, f"Expected 100, got {result}"  
print(f"✓ Sum of odd numbers 1-20 is {result}")
```

Test your solution

```
# Test case: Example from solution  
result = sum_odd_numbers()  
print(f"Sum of odd numbers 1-20: {result}") # 100
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 10: Numbers Divisible by K (1 to N)

Exercise Description

Write a function to print numbers between 1 and N that are divisible by K using a for loop and range.

Your solution

```
def print_divisible_by_k(n, k):  
    # Step 1: Set up for loop with start=k, stop=n+1, step=k  
    # Step 2: For each number divisible by k, print it
```

Test your solution

```
# Test case 2: Test with n=20, k=4  
# Expected: prints 4, 8, 12, 16, 20  
print_divisible_by_k(20, 4)
```

Test your solution

```
# Test case: Example from solution  
print_divisible_by_k(20, 3) # Prints 3, 6, 9, 12, 15, 18
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 11: Count Digits in a Number

Exercise Description

Write a function that counts how many digits are in a positive integer.

Your solution

```
def count_digits(num):  
    # Step 1: Initialize a counter to 0  
    # Step 2: While num > 0, do integer division by 10 and increment counter  
    # Step 3: Return the counter
```

Test your solution

```
assert count_digits(123456) == 6  
assert count_digits(7) == 1  
assert count_digits(10) == 2  
print("✓ count_digits tests passed")
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 12: Cubes from 1 to 5

Exercise Description

Write a function that returns a list with the cubes of numbers from 1 to 5 using a for loop.

Your solution

```
def cubes_1_to_5():  
    # Step 1: Initialize an empty list  
    # Step 2: For i from 1 to 5 inclusive, append i**3 to the list  
    # Step 3: Return the list
```

Test your solution

```
assert cubes_1_to_5() == [1, 8, 27, 64, 125]  
print("✓ cubes_1_to_5 tests passed")
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 13: Product of Numbers from 1 to N

Exercise Description

Write a function that returns the product of numbers from 1 to N using a for loop.

Your solution

```
def product_1_to_n(n):  
    # Step 1: Initialize product to 1  
    # Step 2: For i from 1 to n inclusive, multiply product by i  
    # Step 3: Return product
```

Test your solution

```
assert product_1_to_n(1) == 1  
assert product_1_to_n(3) == 6  
assert product_1_to_n(5) == 120  
print("✓ product_1_to_n tests passed")
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 14: Odd Numbers from 1 to 15

Exercise Description

Write a function that prints all odd numbers from 1 to 15 using a while loop.

Your solution

```
def odd_1_to_15():  
    # Step 1: Initialize i to 1  
    # Step 2: While i <= 15, print i then increment by 2
```

Test your solution

```
import io  
import sys
```



```

# Capture the printed output
captured_output = io.StringIO()
sys.stdout = captured_output

odd_1_to_15()

# Restore normal output
sys.stdout = sys.__stdout__

# Get the output and convert to list of integers for testing
output_lines = captured_output.getvalue().strip().split('\n')
result = [int(line.strip()) for line in output_lines if line.strip()]

assert result == [1, 3, 5, 7, 9, 11, 13, 15]
print("✓ odd_1_to_15 tests passed")
print(f"Generated odd numbers: {result}")

```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```

from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth

```

Exercise 15: Print Numbers from 5 to 1

Exercise Description

Write a function that prints numbers from 5 to 1 using a for loop with range.

Your solution

```
def countdown_5_to_1():  
    # Step 1: Use range from 5 down to 1 inclusive and print each number
```

Test your solution

```
import io  
import sys  
  
# Capture the printed output  
captured_output = io.StringIO()  
sys.stdout = captured_output  
  
countdown_5_to_1()  
  
# Restore normal output  
sys.stdout = sys.__stdout__  
  
# Get the output and convert to list of integers for testing  
output_lines = captured_output.getvalue().strip().split('\n')  
result = [int(line.strip()) for line in output_lines if line.strip()]  
  
assert result == [5, 4, 3, 2, 1]
```

```
print("✓ countdown_5_to_1 tests passed")
print(f"Generated countdown: {result}")
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 16: Sum of the First 7 Odd Numbers

Exercise Description

Write a function to find the sum of the first 7 odd numbers using a while loop.

Your solution

```
def sum_first_7_odds():
    # Step 1: Initialize i=1, count=0, total=0
    # Step 2: While count < 7, add i to total, increment i by 2 and count by 1
    # Step 3: Return total
```

Test your solution

```
assert sum_first_7_odds() == 49
print("✓ sum_first_7_odds tests passed")
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 17: Repeat String as Many Times as Its Length

Exercise Description

Write a function that takes a string and returns a concatenated string with the original string repeated as many times as its length. Use a loop and remember that `len(s)` gives you the length of string `s`.

Your solution

